SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5 Total Marks: 25

Code of Conduct
I THIRU KUMARAN S 192521149 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thiru Kumaran S

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

① Cloud Auto - Scaling and Load Balancing

Auto - scaling automatically increases or decreases cloud resources based on traffic demand. Load balancing distributes incoming request across multiple servers to avoid overloading a single server.

* Auto - scaling handles sudden traffic spikes during seasonal sales.

* Load balancing distributes request evenly among server

* multiple servers reduce downtime if one server fail

* monitoring tools track CPU usage, traffic, reponse time, and errors.

Real - world example: During an Amazon great Indian Festival sale, auto - scaling add servers when customer traffic increases while load balancing distributes requests.

② Cloud storage, Security and Disaster Recovery.

Cloud storage stores large amount of data on remote servers and allow access through the internet

⇒ Durability: Data is stored using reliable storage system to prevent loss

⇒ Redundancy: multiple copies of data are maintained

different devices or locations.

⇒ Encryption: Data is protected during storage and transmission

⇒ Access control: Only authorized used can access sensitive backup data.

Real world example - A bank stores encrypted backup in Aws s3 If its local data centre fails, if can restore the backup from another region and continue it service.

③ Content Delivery Networking (CDNs) and global Availability

A content Delivery Network (CDN) is a group of distributed servers that delivers content to user from locations near them.

⇒ Edge server store cached copies of videos, images, and other content.

⇒ Nearby servers reduce latency and loading time.

⇒ CDNs reduce the load on the main servers.

⇒ Traffic can be redirected to healthy servers during failures.

Real-world Example: Netflix uses content delivery infrastructures to deliver videos efficiently user can stream video with less buffering because content is served from nearby location.

④ Cloud Security, Compliance and Data Sovereignty.

Cloud Providers offer security tools to protect sensitive financial information and support regulatory compliance.

⇒ Compliance - Providers offer service and certification that help organization meeting applicable regulation.

⇒ Encryption - Data is encrypted at rest and in transit

⇒ Access Control: Multi factor authentication and role-based access restrict authorized access.

⇒ Logging - Recording user activities transactions and system changes.

Real-world Example - A bank stores customer transaction data in an approved Indian cloud region, encrypts the data restrict access to authorized employees and maintain audit logs to support compliance.

⑤ Container, Virtual Machines and Kubernates.

Containers package an application with its libraries and dependences Virtual machines (VMs) run seperate Operating System on virtualized hardware.

Containers	Virtual machines
Share the host OS kernal	Run separate guest operating system
Leightweight	Generally heavier
Start quickly	Usually taken longer to start
Use fewer resources	Use more resources